VIVEK KUMAR

Data Engineer • PySpark & Apache Kafka • AWS Big Data • Medallion Architecture • GenAI

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with demonstrated expertise in architecting batch and streaming data pipelines, distributed data processing with PySpark, and cloud data lakes on AWS. Proficient in Medallion Architecture, Kafka event streaming, and GenAI-powered conversational data analytics (Text-to-SQL & RAG). Dual AWS Certified: Data Engineer – Associate & Developer – Associate.

TECHNICAL SKILLS

Big Data & Streaming: Apache Spark (PySpark), Apache Kafka, PySpark Structured Streaming, Delta Lake, Airflow

Data Modeling & Arch: Medallion Architecture (Bronze/Silver/Gold), Star/Snowflake Schema, Data Warehousing, Data Lakehouse

AWS & Cloud Data: AWS (S3, Glue, Athena, EMR, Redshift, RDS PostgreSQL, CloudWatch), Databricks, Docker

GenAI & ML Analytics: LangChain, Text-to-SQL, Vector Search, ChromaDB, PyTorch (CNNs, ResNet-18), OpenCV

Languages & DBs: Python, SQL (PostgreSQL, MySQL), Java, Bash, MongoDB, Redis

BI & Analytics: Power BI, Tableau, Metabase, Matplotlib, Seaborn, Query Optimization

PROFESSIONAL EXPERIENCE

Data & Backend Engineering Intern | HulkHire (Remote) *Jul 2024 – Dec 2024*

- Designed and optimized SQL schemas in AWS RDS PostgreSQL for high-volume transaction processing, improving query execution by 35%.
- Built streaming event pipelines using Apache Kafka and ActiveMQ to ingest transaction telemetry for reconciliation and audit logging.
- Developed Python data transformation and sanitization scripts, eliminating invalid transaction payloads before persistence.
- Integrated Redis caching to reduce repeated analytical query hits on transactional databases by 40%.
- Documented data dictionaries, pipeline schemas, and collaborated across cross-functional engineering teams using Jira and Git.

KEY PROJECTS

Real-Time Banking Fraud Detection Streaming Pipeline | Apache Kafka, PySpark Structured Streaming, Docker, Medallion Architecture *GitHub*

- Constructed streaming fraud detection pipeline ingesting transaction streams via Kafka and processing with PySpark Structured Streaming.
- Implemented Medallion Architecture: Bronze (raw JSON ingestion to Parquet), Silver (schema validation & deduplication), Gold (real-time velocity scoring).
- Deployed multi-container cluster via Docker Compose, enabling sub-second detection and alert dispatch on high-risk transactions.

Batch Sales ETL & Data Warehouse Pipeline | PySpark, AWS S3, PostgreSQL, Delta Lake, Power BI, Python *GitHub*

- Engineered distributed batch ETL pipeline extracting multi-source POS logs and staging clean star-schema datasets into PostgreSQL/AWS S3.
- Automated dirty data handling (duplicate transactions, missing customer keys, negative balances) with quarantine error-logging.
- Configured interactive Power BI reporting dashboards connected to analytical views for real-time sales trend forecasting.

GenAI Text-to-SQL Analytics Assistant | LangChain, LCEL, OpenAI API, PostgreSQL, ChromaDB *GitHub*

- Developed an AI assistant converting natural language business questions into precise, read-only SQL queries against warehouse schemas.
- Leveraged vector retrieval over database schema metadata and few-shot prompt chaining to ensure zero hallucination and high execution accuracy.

Deepfake Presentation Attack Detection (PAD) API | PyTorch, ResNet-18, OpenCV, FastAPI, Docker *GitHub*

- Trained a computer vision ResNet-18 deep neural network to detect biometric facial presentation attacks (print, replay spoofing).
- Packaged model into a low-latency FastAPI inference service with Docker containerization.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering | DIT University, Dehradun *2021 – 2025*

Certifications: AWS Certified Developer – Associate (DVA-C02) • AWS Certified Data Engineer – Associate (DEA-C01) • Node.js & Express (Coursera) • React Frontend Development (freeCodeCamp) • Agile Project Management (LinkedIn Learning)